

Please check that this question paper contains 09 questions and 02 printed pages within first ten minutes.

MORNING

[Total No. of Questions: 09]

01 MAR 2024

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Uni. Roll No.

Program: B.Tech. (IT)

Semester: 3rd

Name of Subject: Object Oriented Programming using C++

Subject Code: PCIT-102

Paper ID: 16041

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1.

- a) Why do we need the preprocessor directive #include?
- b) Distinguish between class and structure?
- c) What is the function parameter? Difference between parameter and argument.
- d) Outline the purpose of scope resolution operator.
- e) Point out the need of super keyword.
- f) Differentiate between virtual functions and pure virtual functions.

Part – B

[Marks: 04 each]

- Q2. Write a program to demonstrate the use of unary, binary and special operators in C++.
- Q3. What is operator overloading? Write a c++ program to overload new operator.
- Q4. What's the difference between public, private, and protected?
- Q5. Write a C++ program to check the given integer is prime or composite number.
- Q6. What is the ambiguity that arises in multiple inheritances? How it can be overcome. Explain with example.
- Q7. Describe the use of following manipulators. Also describe the need of pointers.

- Q8. (i) Define a function template giving its syntax. Write a c++ program to implement array representation of a stack for integers, characters and floating point numbers using class template.
- (ii) Explain try catch and throw exception handling in c++.

OR

- (i) Write a C++ program demonstrating use of the pure virtual function with the use of base and derived classes
- (ii) What does inheritance means in c++? What are different forms of inheritance? Give an example of each.

- Q9. (i) Write a program in C++ that display entered string into reverse order
- (ii) What is a friend function? What are its merits and demerits?

OR

- (i) Write a c++ program to perform multiplication between an integer and complex number object using friend operator function.
- (ii) Create class called shopping list. The shopping list includes details such as the item code number and price of item. Provide operations such as adding an item to the list, deleting an item from the list printing the total value.
